Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: AJR Farms (Formerly Maria Silveira 2011 Trust Dairy)

Physical address of dairy:

Number and Street	City	County	Zip Code
10511 Ingomar Grade RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 02/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X080-X051-XXXX

B. OPERATORS

Farms, AJR

Operator name: Farms, AJR Telephone no.: (209) 854-0968 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
10511 Ingomar Grade	Gustine	CA	95322

This operator is responsible for paying permit fees.

Maria C. Silveira 2011 Trust

Operator name: Maria C. Silveira 2011 Trust Telephone no.: (209) 854-0968 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
10511 Ingomar Grade RD	Gustine	CA	95322

Silveira, John

Operator name: Silveira, John Telephone no.: (209) 826-0968 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
10511 Ingomar Grade RD	Gustine	CA	95322

C. OWNERS

Farms, AJR

Legal owner name: Farms, AJR

Telephone no.: (209) 854-0968

Landline	Cellular

Mailing Address Number and Street	City	State	Zip Code
10511 Ingomar Grade	Gustine	CA	95322

This owner is responsible for paying permit fees.

Maria C. Silveira 2011 Trust

Legal owner name: Maria C. Silveira 2011 Trust

Telephone no.: (209) 854-0968

Landline	Cellular

Mailing Address Number and Street	City	State	Zip Code
10511 Ingomar Grade RD	Gustine	CA	95322

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	26	72	60	65	45
Number under roof	203	0	0	0	0	0
Maximum number	203	26	72	60	65	45
Average number	203	26	72	60	65	45
Avg live weight (lbs)	1,400	1,500	900	750		

Predominant milk cow breed: Holstein

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 7,041.02 *tons per reporting period*

Total nitrogen from manure: 87,860.01 *lbs per reporting period*

After ammonia losses (30% loss applied): 61,502.01 *lbs per reporting period*

Total phosphorus from manure: 14,096.89 *lbs per reporting period*

Total potassium from manure: 37,542.66 *lbs per reporting period*

Total salt from manure: 101,561.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 11,268,000 *gallons*

Total nitrogen generated: 13,806.73 *lbs*

Total phosphorus generated: 5,666.78 *lbs*

Total potassium generated: 49,416.99 *lbs*

Total salt generated: 306,128.48 *lbs*

	11,268,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	11,268,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Domestic Well	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Cement Ditch	16	16	2	process wastewater	X070-X080-X051-XXXX
Corral Field	16	16	2	both	X070-X080-X051-XXXX
Little Pasture	12	12	6	manure	X070-X080-X051-XXXX
Middle Field	18	18	2	both	X070-X080-X051-XXXX
The 12	12	12	2	process wastewater	X070-X080-X051-XXXX
The 28	28	28	0	none	X070-X080-X032-XXXX
The 30	30	30	2	both	X070-X080-X052-XXXX
The Swamp	35	35	6	manure	X070-X080-X052-XXXX
Totals for areas that were used for application	139	139	22		
Totals for areas that were not used for application	28	28	0		
Land application area totals	167	167	22		

B. CROPS AND HARVESTS

Cement Ditch

Field name: Cement Ditch

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	165.00 *ton*	Dry-weight		73.1	17,800.00	4,000.00	31,300.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.31	98.76	22.19	173.66	754.55

Cement Ditch

07/28/2023: Sorghum-Sudangrass, forage

Crop: Sorghum-Sudangrass, forage Acres planted: 16 Plant date: 07/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	155.76 *ton*	Dry-weight		71.4	14,800.00	2,500.00	26,400.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	9.74	82.41	13.92	147.01	612.53

Corral Field

Field name: Corral Field

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	151.80 *ton*	Dry-weight		71.2	17,200.00	3,600.00	30,900.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.49	93.99	19.67	168.86	743.21

07/28/2023: Sorghum-Sudangrass, forage

Crop: Sorghum-Sudangrass, forage Acres planted: 16 Plant date: 07/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	167.64 *ton*	Dry-weight		72.1	16,100.00	2,800.00	27,200.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	256.00	35.20	264.00	0.00
Total actual harvest content	10.48	94.13	16.37	159.02	666.49

Little Pasture

Field name: Little Pasture

09/13/2007: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 12 Plant date: 09/13/2007

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
03/25/2023	12.21 *ton*	Dry-weight		72.3	28,500.00	3,800.00	28,100.00		10.80
04/30/2023	11.88 *ton*	Dry-weight		68.4	19,200.00	2,200.00	32,700.00		13.00
05/28/2023	12.21 *ton*	Dry-weight		68.4	19,200.00	2,200.00	32,700.00		13.00
06/30/2023	12.54 *ton*	Dry-weight		68.4	19,200.00	2,200.00	32,700.00		13.00
08/05/2023	12.41 *ton*	Dry-weight		72.3	28,500.00	3,800.00	28,100.00		10.80
09/20/2023	12.28 *ton*	Dry-weight		72.3	28,500.00	3,800.00	28,100.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	6.13	85.59	10.72	110.95	434.78

Middle Field

Field name: Middle Field

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	217.80 *ton*	Dry-weight		71.2	17,200.00	3,600.00	30,900.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.10	119.88	25.09	215.36	947.87

Middle Field

07/28/2023: Sorghum-Sudangrass, forage

Crop: Sorghum-Sudangrass, forage Acres planted: 18 Plant date: 07/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	187.44 *ton*	Dry-weight		71.4	14,800.00	2,500.00	26,400.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	256.00	35.20	264.00	0.00
Total actual harvest content	10.41	88.16	14.89	157.25	655.21

The 12

Field name: The 12

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	153.12 *ton*	Dry-weight		73.1	17,800.00	4,000.00	31,300.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.76	122.19	27.46	214.87	933.62

07/28/2023: Sorghum-Sudangrass, forage

Crop: Sorghum-Sudangrass, forage Acres planted: 12 Plant date: 07/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	102.96 *ton*	Dry-weight		71.4	14,800.00	2,500.00	26,400.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	256.00	35.20	264.00	0.00
Total actual harvest content	8.58	72.63	12.27	129.56	539.85

The 30

Field name: The 30

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	300.96 *ton*	Dry-weight		72.5	16,300.00	3,800.00	30,200.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.03	89.94	20.97	166.63	684.18

07/28/2023: Sorghum-Sudangrass, forage

Crop: Sorghum-Sudangrass, forage Acres planted: 30 Plant date: 07/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	320.76 *ton*	Dry-weight		72.0	15,900.00	2,700.00	26,700.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	256.00	35.20	264.00	0.00
Total actual harvest content	10.69	95.20	16.17	159.87	652.64

The Swamp

Field name: The Swamp

The Swamp

09/13/2007: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 35 Plant date: 09/13/2007

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
03/25/2023	35.64 *ton*	Dry-weight		73.5	36,900.00	4,500.00	35,600.00		10.20
04/30/2023	36.96 *ton*	Dry-weight		68.3	19,800.00	2,300.00	33,800.00		13.30
05/28/2023	34.32 *ton*	Dry-weight		68.3	19,800.00	2,300.00	33,800.00		13.30
06/30/2023	35.64 *ton*	Dry-weight		68.3	19,800.00	2,300.00	33,800.00		13.30
08/05/2023	36.96 *ton*	Dry-weight		68.3	19,800.00	2,300.00	33,800.00		13.30
09/20/2023	36.96 *ton*	Dry-weight		68.3	19,800.00	2,300.00	33,800.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	6.19	84.78	9.96	129.93	490.73

NUTRIENT BUDGET

A. LAND APPLICATIONS

Cement Ditch - 11/26/2022: Oats, silage-soft dough

Field name: Cement Ditch

Crop: Oats, silage-soft dough — Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.56	30.71	296.87	1,688.80	726,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	185.28	2,065,400.00 *gal*
Application event totals		108.56	30.71	296.87	1,874.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	194.15	2,164,180.00 *gal*
Application event totals		0.00	0.00	0.00	194.15	

Cement Ditch - 07/28/2023: Sorghum-Sudangrass, forage

Field name: Cement Ditch

Crop: Sorghum-Sudangrass, forage — Plant date: 07/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.13	17.60	150.27	947.64	588,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	177.23	1,975,600.00 *gal*
Application event totals		39.13	17.60	150.27	1,124.87	

Cement Ditch - 07/28/2023: Sorghum-Sudangrass, forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.53	17.78	151.81	957.31	594,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	197.37	2,200,100.00 *gal*
Application event totals		39.53	17.78	151.81	1,154.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	178.84	1,993,560.00 *gal*
Application event totals		0.00	0.00	0.00	178.84	

Corral Field - 11/26/2022: Oats, silage-soft dough

Field name: Corral Field

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.50	26.76	228.48	1,440.80	894,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	203.81	2,271,940.00 *gal*
Application event totals		59.50	26.76	228.48	1,644.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	204.62	2,280,920.00 *gal*
Application event totals		0.00	0.00	0.00	204.62	

Corral Field - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	58.21	12.63	53.59	0.00	35.00 *ton*
Application event totals		58.21	12.63	53.59	0.00	

Corral Field - 07/28/2023: Sorghum-Sudangrass, forage

Field name: Corral Field

Crop: Sorghum-Sudangrass, forage

Plant date: 07/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	49.90	10.82	45.94	0.00	30.00 *ton*
Application event totals		49.90	10.82	45.94	0.00	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.94	15.27	130.34	821.93	510,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	176.42	1,966,620.00 *gal*
Application event totals		33.94	15.27	130.34	998.35	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.13	17.60	150.27	947.64	588,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	194.15	2,164,180.00 *gal*
Application event totals		39.13	17.60	150.27	1,141.78	

Corral Field - 07/28/2023: Sorghum-Sudangrass, forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	174.81	1,948,660.00 *gal*
Application event totals		0.00	0.00	0.00	174.81	

Little Pasture - 09/13/2007: Clover-grass, hay

Field name: Little Pasture

Crop: Clover-grass, hay

Plant date: 09/13/2007

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	30.08	251,440.00 *gal*
Application event totals		0.00	0.00	0.00	30.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.00	242,460.00 *gal*
Application event totals		0.00	0.00	0.00	29.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	32.22	269,400.00 *gal*
Application event totals		0.00	0.00	0.00	32.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	32.22	269,400.00 *gal*
Application event totals		0.00	0.00	0.00	32.22	

Little Pasture - 09/13/2007: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.30	278,380.00 *gal*
Application event totals		0.00	0.00	0.00	33.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.15	260,420.00 *gal*
Application event totals		0.00	0.00	0.00	31.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	77.62	16.84	71.46	0.00	35.00 *ton*
Application event totals		77.62	16.84	71.46	0.00	

Middle Field - 11/26/2022: Oats, silage-soft dough

Field name: Middle Field

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.50	30.82	263.06	1,658.90	1,158,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	181.17	2,271,940.00 *gal*
Application event totals		68.50	30.82	263.06	1,840.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	189.76	2,379,700.00 *gal*
Application event totals		0.00	0.00	0.00	189.76	

Middle Field - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	73.92	16.04	68.05	0.00	50.00 *ton*
Application event totals		73.92	16.04	68.05	0.00	

Middle Field - 07/28/2023: Sorghum-Sudangrass, forage

Field name: Middle Field

Crop: Sorghum-Sudangrass, forage

Plant date: 07/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	44.35	9.62	40.83	0.00	30.00 *ton*
Application event totals		44.35	9.62	40.83	0.00	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.43	15.49	132.21	833.75	582,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	173.29	2,173,160.00 *gal*
Application event totals		34.43	15.49	132.21	1,007.04	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.88	13.89	118.58	747.80	522,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	191.91	2,406,640.00 *gal*
Application event totals		30.88	13.89	118.58	939.70	

Middle Field - 07/28/2023: Sorghum-Sudangrass, forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	174.01	2,182,140.00 *gal*
Application event totals		0.00	0.00	0.00	174.01	

The 12 - 11/26/2022: Oats, silage-soft dough

Field name: The 12

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	126.80	35.87	346.75	1,972.59	636,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	207.30	1,733,140.00 *gal*
Application event totals		126.80	35.87	346.75	2,179.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	208.38	1,742,120.00 *gal*
Application event totals		0.00	0.00	0.00	208.38	

The 12 - 07/28/2023: Sorghum-Sudangrass, forage

Field name: The 12

Crop: Sorghum-Sudangrass, forage

Plant date: 07/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

The 12 - 07/28/2023: Sorghum-Sudangrass, forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.87	17.48	149.25	941.19	438,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	183.67	1,535,580.00 *gal*
Application event totals		38.87	17.48	149.25	1,124.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.87	17.48	149.25	941.19	438,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	197.64	1,652,320.00 *gal*
Application event totals		38.87	17.48	149.25	1,138.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	185.82	1,553,540.00 *gal*
Application event totals		0.00	0.00	0.00	185.82	

The 30 - 11/26/2022: Oats, silage-soft dough

Field name: The 30

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.06	27.02	230.62	1,454.33	1,692,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	192.05	4,014,060.00 *gal*
Application event totals		60.06	27.02	230.62	1,646.38	

The 30 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	186.04	3,888,340.00 *gal*
Application event totals		0.00	0.00	0.00	186.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	44.35	9.62	40.83	0.00	50.00 *ton*
Application event totals		44.35	9.62	40.83	0.00	

The 30 - 07/28/2023: Sorghum-Sudangrass, forage

Field name: The 30

Crop: Sorghum-Sudangrass, forage

Plant date: 07/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	44.35	9.62	40.83	0.00	50.00 *ton*
Application event totals		44.35	9.62	40.83	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.86	15.23	130.03	820.00	954,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	175.72	3,672,820.00 *gal*
Application event totals		33.86	15.23	130.03	995.72	

The 30 - 07/28/2023: Sorghum-Sudangrass, forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.65	15.14	129.21	814.84	948,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	186.47	3,897,320.00 *gal*
Application event totals		33.65	15.14	129.21	1,001.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	176.58	3,690,780.00 *gal*
Application event totals		0.00	0.00	0.00	176.58	

The Swamp - 09/13/2007: Clover-grass, hay

Field name: The Swamp

Crop: Clover-grass, hay

Plant date: 09/13/2007

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.83	898,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.09	880,040.00 *gal*
Application event totals		0.00	0.00	0.00	36.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.35	862,080.00 *gal*
Application event totals		0.00	0.00	0.00	35.35	

The Swamp - 09/13/2007: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.35	862,080.00 *gal*
Application event totals		0.00	0.00	0.00	35.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.09	880,040.00 *gal*
Application event totals		0.00	0.00	0.00	36.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.46	889,020.00 *gal*
Application event totals		0.00	0.00	0.00	36.46	

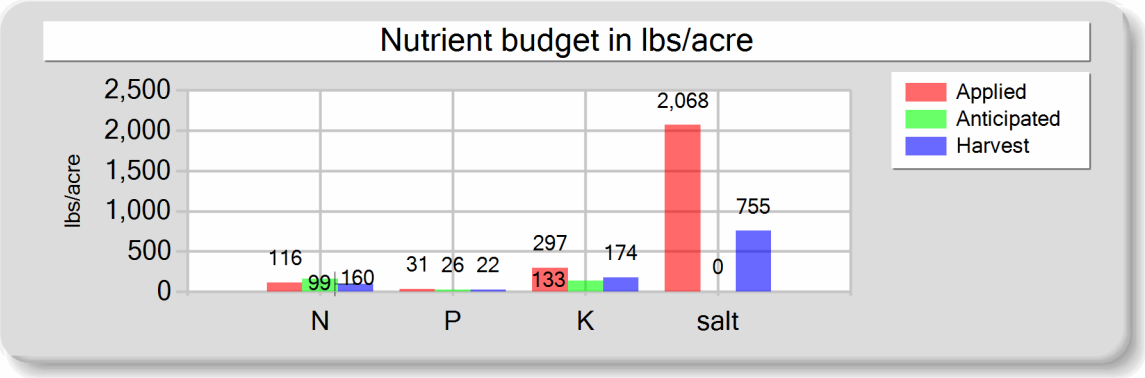
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	57.02	12.37	52.50	0.00	75.00 *ton*
Application event totals		57.02	12.37	52.50	0.00	

B. NUTRIENT BUDGET

Cement Ditch - 11/26/2022: Oats, silage-soft dough

Field name: Cement Ditch Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	108.56	30.71	296.87	1,688.80
Fresh water	0.00	0.00	0.00	379.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	115.56	30.71	296.87	2,068.23
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	98.76	22.19	173.66	754.55
Nutrient balance	16.80	8.52	123.21	1,313.68
Applied to removed ratio	1.17	1.38	1.71	2.74

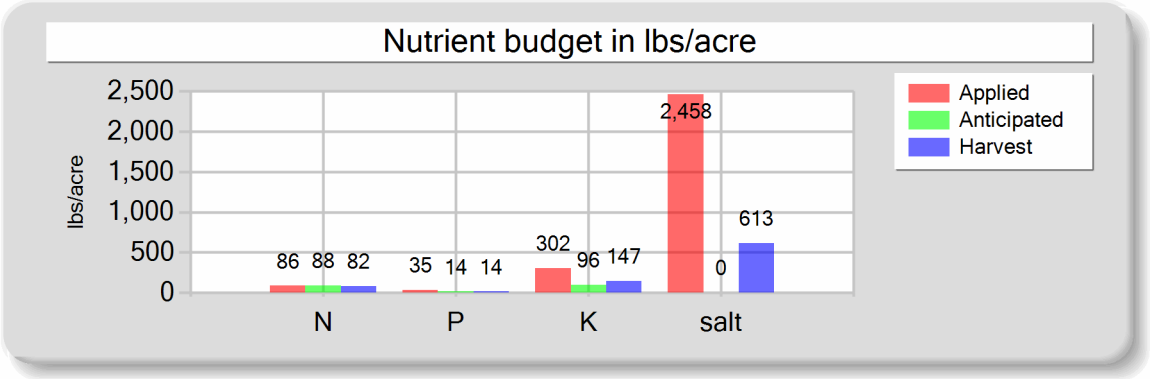
Fresh water applied	
4,229,580.00	*gallons*
155.76	*acre-inches*
9.74	*inches/acre*

Process wastewater applied	
726,000.00	*gallons*
26.74	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Cement Ditch - 07/28/2023: Sorghum-Sudangrass, forage

Field name: Cement Ditch Crop: Sorghum-Sudangrass, forage Plant date: 07/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	78.66	35.39	302.08	1,904.94
Fresh water	0.00	0.00	0.00	553.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	85.66	35.39	302.08	2,458.38
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	82.41	13.92	147.01	612.53
Nutrient balance	3.25	21.47	155.07	1,845.85
Applied to removed ratio	1.04	2.54	2.05	4.01

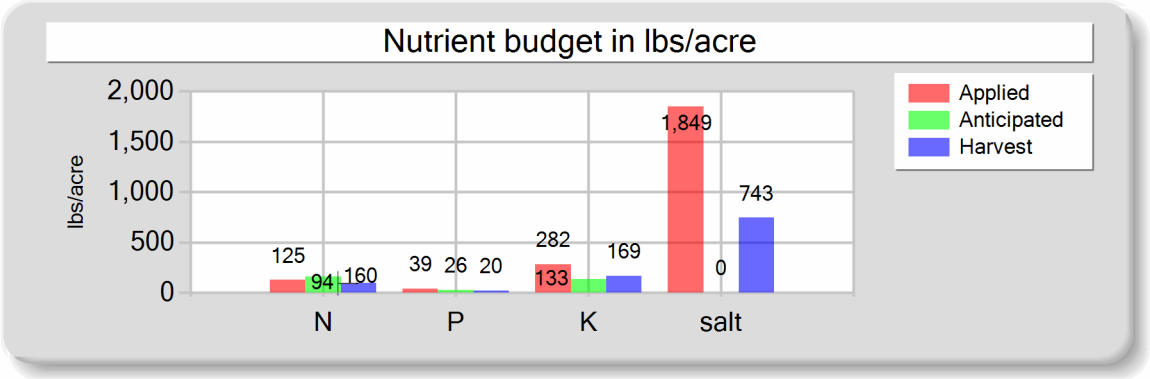
Fresh water applied	
6,169,260.00	*gallons*
227.19	*acre-inches*
14.20	*inches/acre*

Process wastewater applied	
1,182,000.00	*gallons*
43.53	*acre-inches*
2.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

Corral Field - 11/26/2022: Oats, silage-soft dough

Field name: Corral Field | Crop: Oats, silage-soft dough | Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	58.21	12.63	53.59	0.00
Process wastewater	59.50	26.76	228.48	1,440.80
Fresh water	0.00	0.00	0.00	408.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	124.71	39.39	282.07	1,849.23
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	93.99	19.67	168.86	743.21
Nutrient balance	30.71	19.72	113.21	1,106.01
Applied to removed ratio	1.33	2.00	1.67	2.49

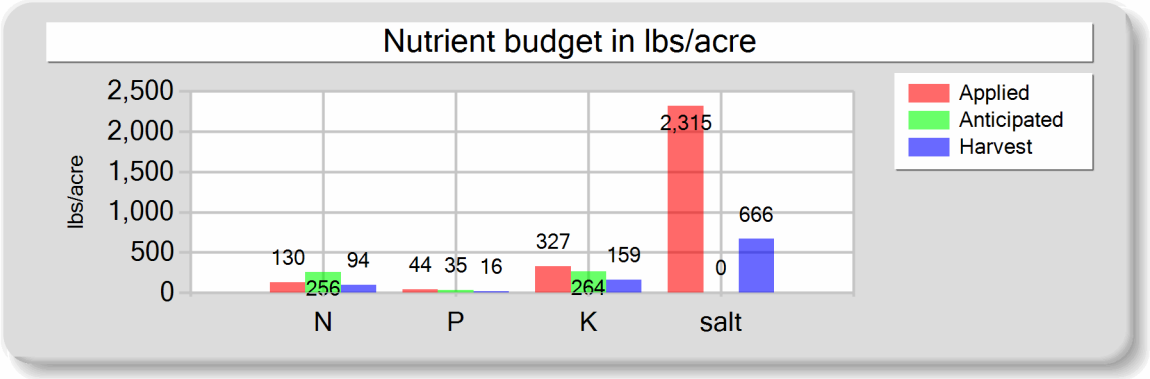
Fresh water applied	
4,552,860.00	*gallons*
167.67	*acre-inches*
10.48	*inches/acre*

Process wastewater applied	
894,000.00	*gallons*
32.92	*acre-inches*
2.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Corral Field - 07/28/2023: Sorghum-Sudangrass, forage

Field name: Corral Field | Crop: Sorghum-Sudangrass, forage | Plant date: 07/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	49.90	10.82	45.94	0.00
Process wastewater	73.07	32.87	280.61	1,769.57
Fresh water	0.00	0.00	0.00	545.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	129.97	43.70	326.55	2,314.95
Anticipated crop nutrient removal	256.00	35.20	264.00	0.00
Actual crop nutrient removal	94.13	16.37	159.02	666.49
Nutrient balance	35.84	27.33	167.52	1,648.45
Applied to removed ratio	1.38	2.67	2.05	3.47

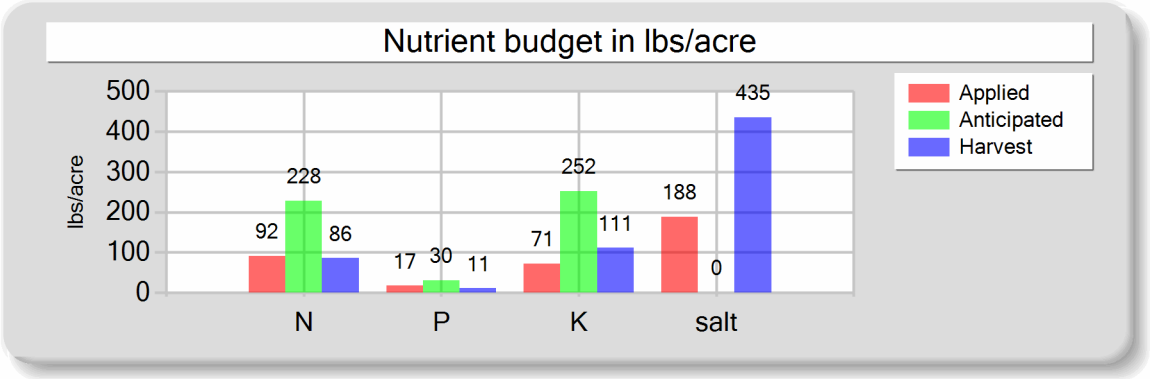
Fresh water applied	
6,079,460.00	*gallons*
223.89	*acre-inches*
13.99	*inches/acre*

Process wastewater applied	
1,098,000.00	*gallons*
40.44	*acre-inches*
2.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

Little Pasture - 09/13/2007: Clover-grass, hay

Field name: Little Pasture | Crop: Clover-grass, hay | Plant date: 09/13/2007



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	77.62	16.84	71.46	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	187.97
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	91.62	16.84	71.46	187.97
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	85.59	10.72	110.95	434.78
Nutrient balance	6.02	6.12	-39.50	-246.81
Applied to removed ratio	1.07	1.57	0.64	0.43

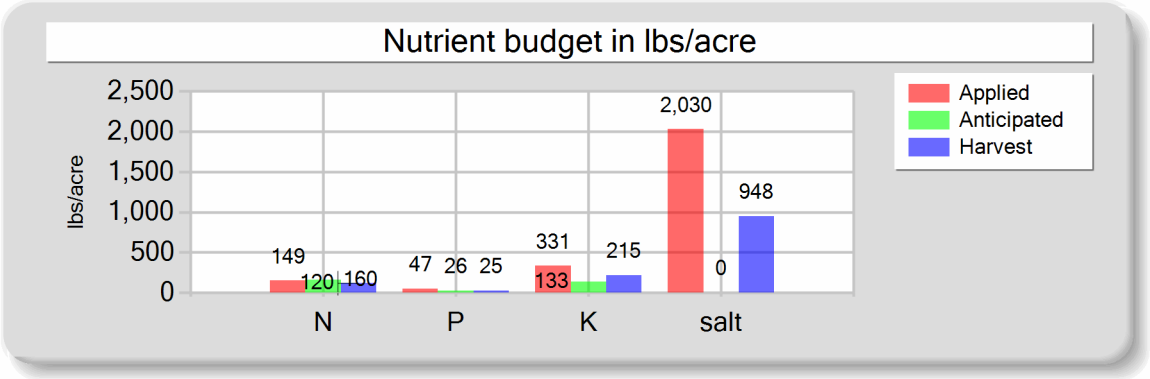
Fresh water applied	
1,571,500.00	*gallons*
57.87	*acre-inches*
4.82	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
6	*harvests*

Middle Field - 11/26/2022: Oats, silage-soft dough

Field name: Middle Field Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	73.92	16.04	68.05	0.00
Process wastewater	68.50	30.82	263.06	1,658.90
Fresh water	0.00	0.00	0.00	370.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	149.42	46.85	331.12	2,029.83
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	119.88	25.09	215.36	947.87
Nutrient balance	29.55	21.76	115.75	1,081.96
Applied to removed ratio	1.25	1.87	1.54	2.14

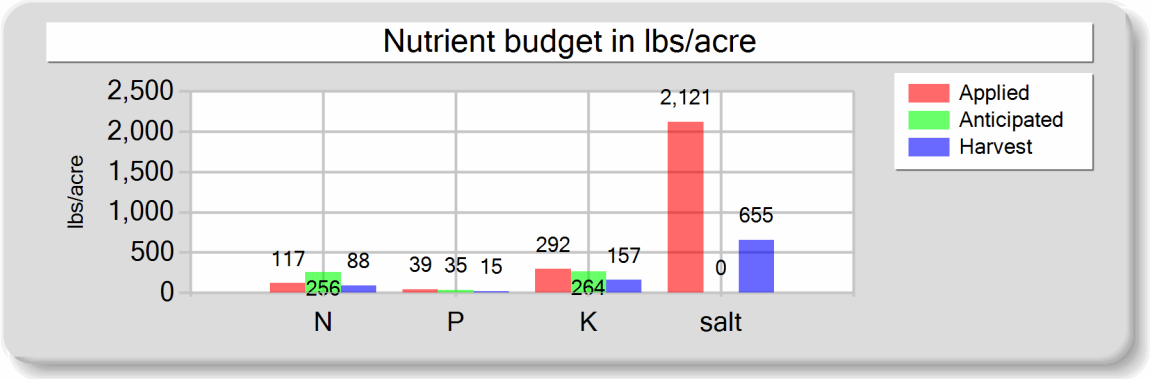
Fresh water applied	
4,651,640.00	*gallons*
171.30	*acre-inches*
9.52	*inches/acre*

Process wastewater applied	
1,158,000.00	*gallons*
42.65	*acre-inches*
2.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Middle Field - 07/28/2023: Sorghum-Sudangrass, forage

Field name: Middle Field | Crop: Sorghum-Sudangrass, forage | Plant date: 07/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	44.35	9.62	40.83	0.00
Process wastewater	65.31	29.38	250.80	1,581.54
Fresh water	0.00	0.00	0.00	539.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	116.66	39.00	291.63	2,120.75
Anticipated crop nutrient removal	256.00	35.20	264.00	0.00
Actual crop nutrient removal	88.16	14.89	157.25	655.21
Nutrient balance	28.51	24.11	134.38	1,465.54
Applied to removed ratio	1.32	2.62	1.85	3.24

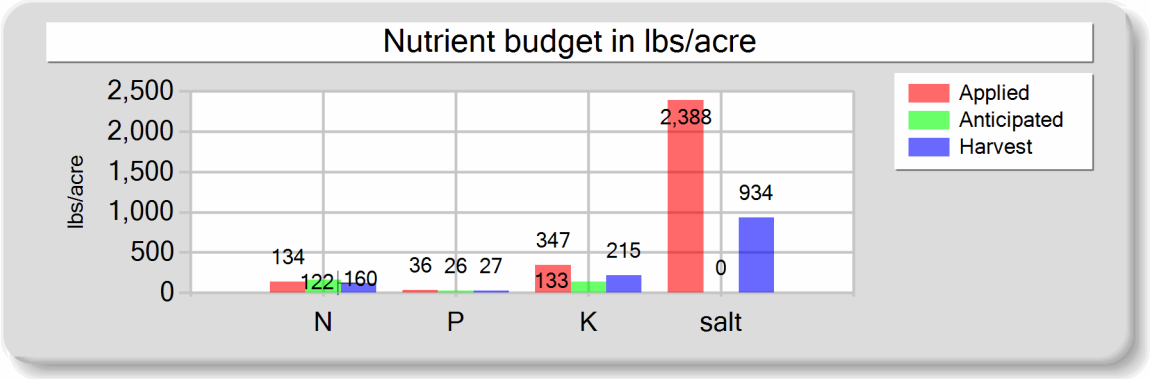
Fresh water applied	
6,761,940.00	*gallons*
249.02	*acre-inches*
13.83	*inches/acre*

Process wastewater applied	
1,104,000.00	*gallons*
40.66	*acre-inches*
2.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 12 - 11/26/2022: Oats, silage-soft dough

Field name: The 12 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	126.80	35.87	346.75	1,972.59
Fresh water	0.00	0.00	0.00	415.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	133.80	35.87	346.75	2,388.27
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	122.19	27.46	214.87	933.62
Nutrient balance	11.61	8.41	131.88	1,454.65
Applied to removed ratio	1.09	1.31	1.61	2.56

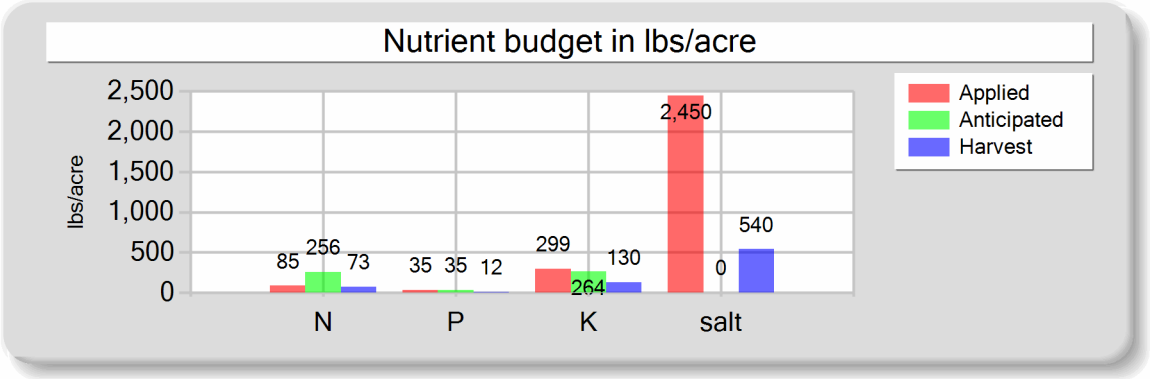
Fresh water applied	
3,475,260.00	*gallons*
127.98	*acre-inches*
10.67	*inches/acre*

Process wastewater applied	
636,000.00	*gallons*
23.42	*acre-inches*
1.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 12 - 07/28/2023: Sorghum-Sudangrass, forage

Field name: The 12 Crop: Sorghum-Sudangrass, forage Plant date: 07/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	77.73	34.97	298.50	1,882.38
Fresh water	0.00	0.00	0.00	567.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	84.73	34.97	298.50	2,449.51
Anticipated crop nutrient removal	256.00	35.20	264.00	0.00
Actual crop nutrient removal	72.63	12.27	129.56	539.85
Nutrient balance	12.10	22.70	168.94	1,909.66
Applied to removed ratio	1.17	2.85	2.30	4.54

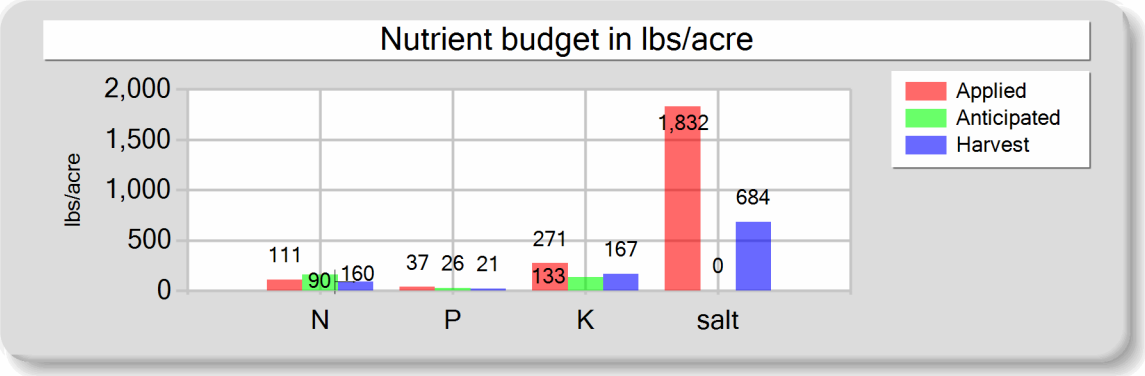
Fresh water applied	
4,741,440.00	*gallons*
174.61	*acre-inches*
14.55	*inches/acre*

Process wastewater applied	
876,000.00	*gallons*
32.26	*acre-inches*
2.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 30 - 11/26/2022: Oats, silage-soft dough

Field name: The 30 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	44.35	9.62	40.83	0.00
Process wastewater	60.06	27.02	230.62	1,454.33
Fresh water	0.00	0.00	0.00	378.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	111.41	36.64	271.45	1,832.42
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	89.94	20.97	166.63	684.18
Nutrient balance	21.47	15.67	104.82	1,148.24
Applied to removed ratio	1.24	1.75	1.63	2.68

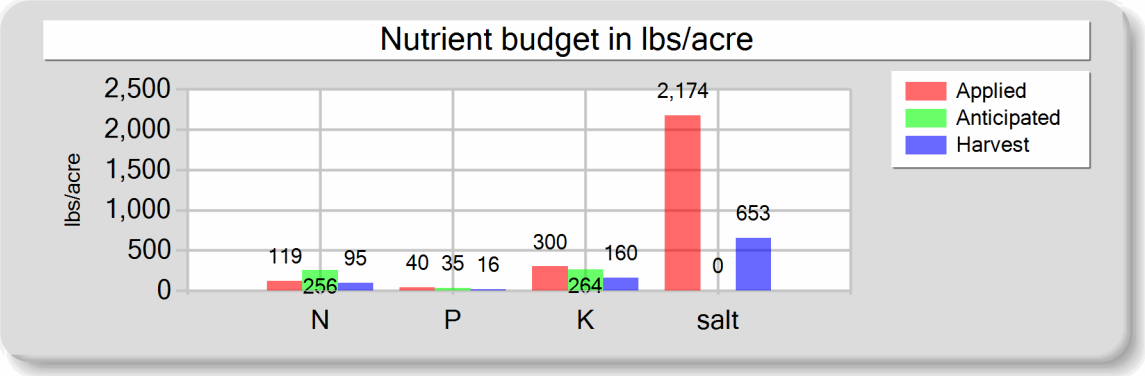
Fresh water applied	
7,902,400.00	*gallons*
291.02	*acre-inches*
9.70	*inches/acre*

Process wastewater applied	
1,692,000.00	*gallons*
62.31	*acre-inches*
2.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 30 - 07/28/2023: Sorghum-Sudangrass, forage

Field name: The 30 Crop: Sorghum-Sudangrass, forage Plant date: 07/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	44.35	9.62	40.83	0.00
Process wastewater	67.51	30.37	259.25	1,634.84
Fresh water	0.00	0.00	0.00	538.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	118.86	39.99	300.08	2,173.61
Anticipated crop nutrient removal	256.00	35.20	264.00	0.00
Actual crop nutrient removal	95.20	16.17	159.87	652.64
Nutrient balance	23.66	23.82	140.21	1,520.97
Applied to removed ratio	1.25	2.47	1.88	3.33

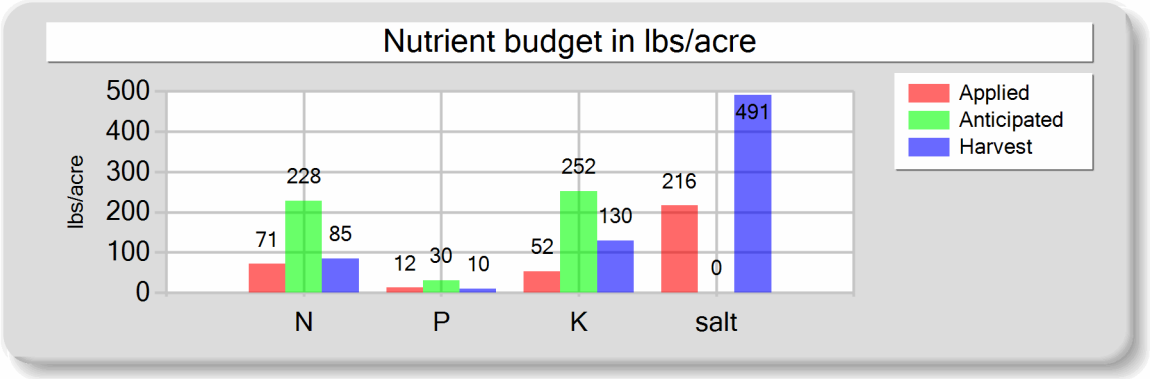
Fresh water applied	
11,260,920.00	*gallons*
414.70	*acre-inches*
13.82	*inches/acre*

Process wastewater applied	
1,902,000.00	*gallons*
70.04	*acre-inches*
2.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

The Swamp - 09/13/2007: Clover-grass, hay

Field name: The Swamp | Crop: Clover-grass, hay | Plant date: 09/13/2007



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	57.02	12.37	52.50	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	216.17
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	71.02	12.37	52.50	216.17
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	84.78	9.96	129.93	490.73
Nutrient balance	-13.75	2.41	-77.44	-274.56
Applied to removed ratio	0.84	1.24	0.40	0.44

Fresh water applied	
5,271,260.00	*gallons*
194.12	*acre-inches*
5.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
6	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure
Sample date: 05/31/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 17.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	6,000.00	42,600.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure
Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 29.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,900.00	4,100.00	17,400.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water
Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	285.00	162.00	0.00	1.70	81.10	784.00								8,080.00	4,460
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	127.00	110.00	0.00	0.60	57.40	490.00								5,890.00	3,090
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	80.90	45.30	0.00	0.50	38.70	328.00								1,950.00	1,550
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Domestic Well

Domestic Well

Sample description: Domestic Well

Sample date: 11/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			30.00								3,270.00	
DL			0.10								10.00	

Irrigation Water - Canal

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										276.00	172
DL	1.00										10.00	10

D. SOIL ANALYSES

Cement Ditch

Ditch

Sample and source description: Ditch

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			141.00				
DL			2.00				

Corral Field

Field

Sample and source description: Field

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			256.00				
DL			2.00				

Little Pasture

Little Pasture

Pasture

Sample and source description: Pasture

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			118.00				
DL			2.00				

Middle Field

Field

Sample and source description: Field

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			144.00				
DL			2.00				

The 12

The 12

Sample and source description: The 12

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			91.00				
DL			2.00				

The 30

The 30

The 30

Sample and source description: The 30

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			256.00				
DL			2.00				

The Swamp

The Swamp

Sample and source description: The Swamp

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Cement Ditch - 11/26/2022: Oats, silage-soft dough

Cement Ditch & The 12

Sample and source description: Cement Ditch & The 12

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,800.00	4,000.00	31,300.00		13.60
DL	100.00	100.00	100.00		0.01

Cement Ditch - 07/28/2023: Sorghum-Sudangrass, forage

Cement Ditch - 07/28/2023: Sorghum-Sudangrass, forage

Cement Ditch & The 12

Sample and source description: Cement Ditch & The 12

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	2,500.00	26,400.00		11.00
DL	100.00	100.00	100.00		0.01

Corral Field - 11/26/2022: Oats, silage-soft dough

Corral Field & Middle Field

Sample and source description: Corral Field & Middle Field

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,600.00	30,900.00		13.60
DL	100.00	100.00	100.00		0.01

Corral Field - 07/28/2023: Sorghum-Sudangrass, forage

Corral Field & Middle Field

Sample and source description: Corral Field & Middle Field

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	2,800.00	27,200.00		11.40
DL	100.00	100.00	100.00		0.01

Little Pasture - 09/13/2007: Clover-grass, hay

Little Pasture

Sample and source description: Little Pasture

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,500.00	3,800.00	28,100.00		10.80
DL	100.00	100.00	100.00		0.01

Little Pasture

Sample and source description: Little Pasture

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,200.00	2,200.00	32,700.00		13.00
DL	100.00	100.00	100.00		0.01

Middle Field - 11/26/2022: Oats, silage-soft dough

Corral Field & Middle Field

Sample and source description: Corral Field & Middle Field

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,600.00	30,900.00		13.60
DL	100.00	100.00	100.00		0.01

The 12 - 11/26/2022: Oats, silage-soft dough

The 12 - 11/26/2022: Oats, silage-soft dough

Cement Ditch & The 12

Sample and source description: Cement Ditch & The 12

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,800.00	4,000.00	31,300.00		13.60
DL	100.00	100.00	100.00		0.01

The 30 - 11/26/2022: Oats, silage-soft dough

The 30

Sample and source description: The 30

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	3,800.00	30,200.00		12.40
DL	100.00	100.00	100.00		0.01

The 30 - 07/28/2023: Sorghum-Sudangrass, forage

The 30

Sample and source description: The 30

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,700.00	26,700.00		10.90
DL	100.00	100.00	100.00		0.01

The Swamp - 09/13/2007: Clover-grass, hay

The Swamp

Sample and source description: The Swamp

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,900.00	4,500.00	35,600.00		10.20
DL	100.00	100.00	100.00		0.01

The Swamp

Sample and source description: The Swamp

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	2,300.00	33,800.00		13.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

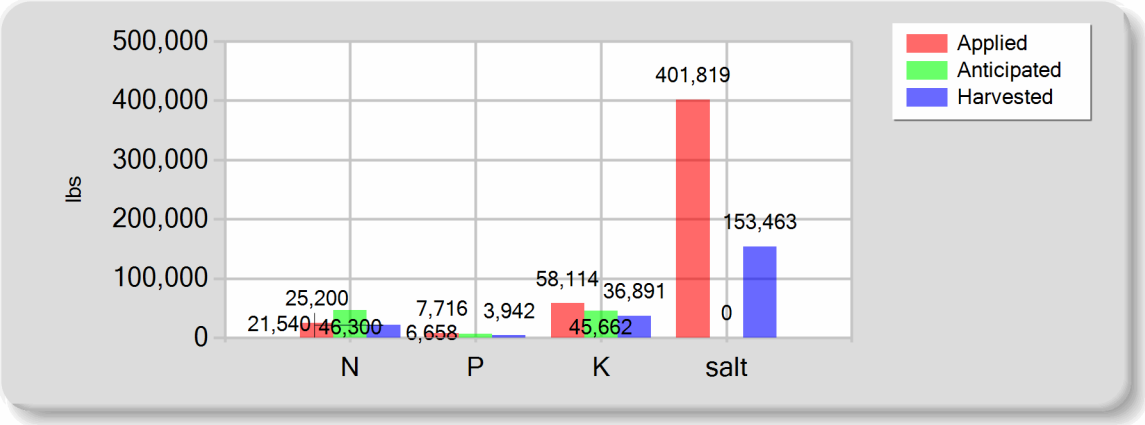
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

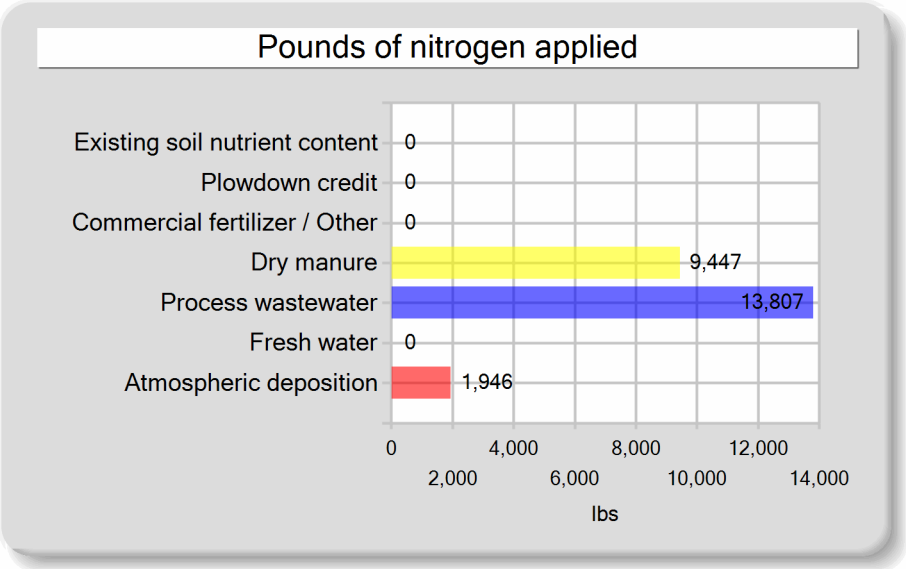
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

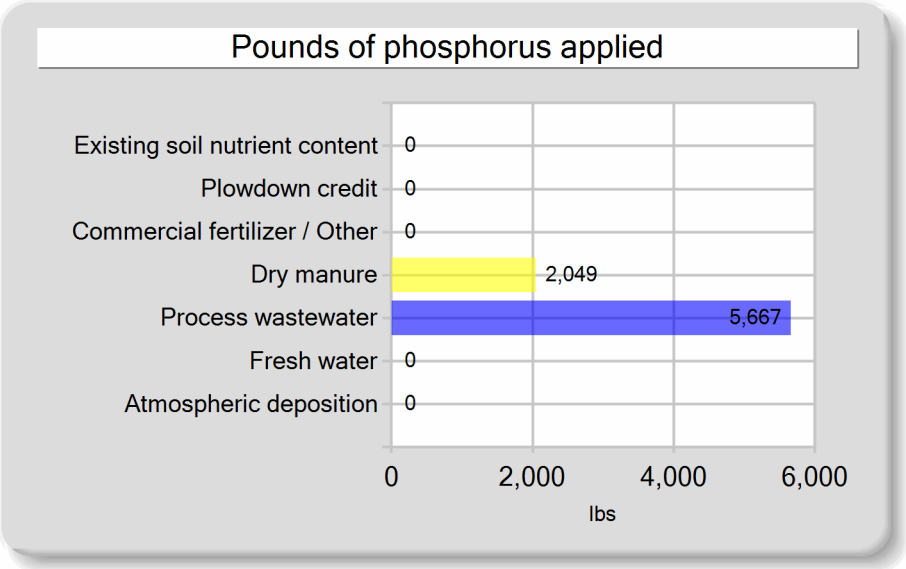
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	9,446.98	2,049.34	8,697.22	0.00
Process wastewater	13,806.73	5,666.78	49,416.99	306,128.48
Fresh water	0.00	0.00	0.00	95,690.56
Atmospheric deposition	1,946.00	0.00	0.00	0.00
Total nutrients applied	25,199.70	7,716.12	58,114.20	401,819.04
Anticipated crop nutrient removal	46,300.00	6,658.00	45,661.60	0.00
Actual crop nutrient removal	21,539.58	3,942.24	36,891.09	153,462.97
Nutrient balance	3,660.12	3,773.89	21,223.11	248,356.07
Applied to removed ratio	1.17	1.96	1.58	2.62

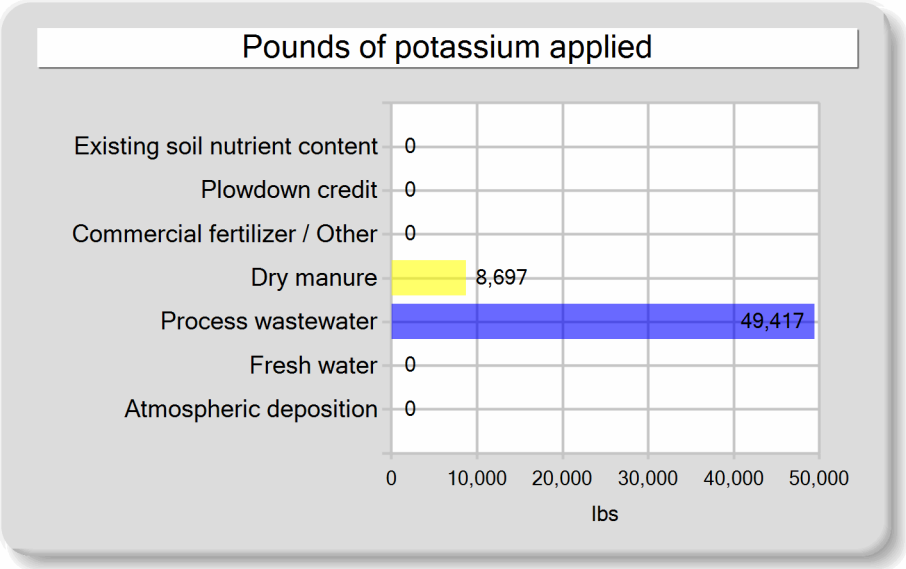
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

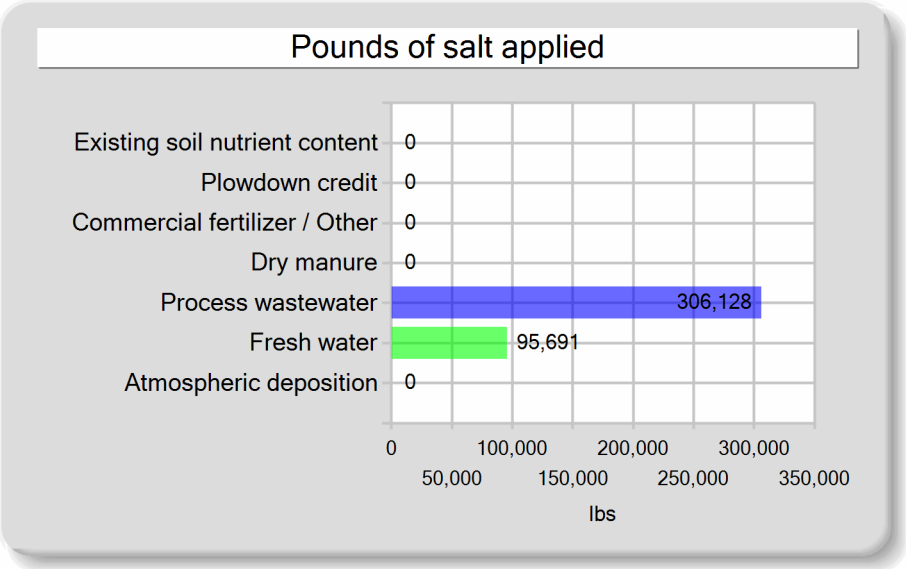


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Clover Grass was harvested as Green Chop and feed directly to the cattle.

No Lagoon Water Sample was taken in the 4th Quarter of 2023 because no process waste water was applied in the 4th Quarter of 2023.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Silveira	John Silveira
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
AJR Farms	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-25-2024	6-25-2024
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: AJR Farms/Maria C. Silveira

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:17

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0483-01	Domestic Well	Ag Water	T. Madruga		11/08/2023 16:00

Default Cooler Temperature on Receipt °C: -0.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: AJR Farms/Maria C. Silveira

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Reported: 11/16/2023 11:17

Sample Results

Sample: Domestic Well
23K0483-01 (Water)

Sampled: 11/8/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**3.27**	mmhos/cm	0.01	1		11/15/23 11:19	SM 2510 B		BEK0350
Electrical Conductivity umhos	**3270**	umhos/cm	10.0	1		11/15/23 11:19	SM 2510 B		BEK0350
Ammonia (as N)	ND	mg/L	0.00	1		11/08/23 16:00	Field		BEK0304
Nitrate Nitrogen as NO3N	**30.0**	mg/L	0.1	1	10	11/10/23 00:34	EPA 300.0		BEK0248
Temperature	**25.0**	units	0.0	1		11/15/23 11:19	SM 4500-H+	H	BEK0350
pH	**7.4**	units	1.0	1		11/15/23 11:19	SM 4500-H+	H	BEK0350

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Received: 11/09/2023 16:20
Reported: 11/16/2023 11:17

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0248										
Blank (BEK0248-BLK1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0248-BLK2)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0248-BLK3)					Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0248-BS1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.1	90-110		
LCS (BEK0248-BS2)					Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.7	90-110		
Duplicate (BEK0248-DUP1)		**Source: 23K0469-01**			Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			1.89	10
Duplicate (BEK0248-DUP2)		**Source: 23K0488-01**			Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	1.0		0.1	mg/L		1.0			0.0995	10
Matrix Spike (BEK0248-MS1)		**Source: 23K0469-01**			Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.2	103	90-110		
Matrix Spike (BEK0248-MS2)		**Source: 23K0488-01**			Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	6.1		0.1	mg/L	5.000	1.0	102	90-110		
Reference (BEK0248-SRM1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0248-SRM2)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0248-SRM3)					Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Ranch: AJR Farms/Maria C. Silveira

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:17

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0350										
Blank (BEK0350-BLK1)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0350-BLK2)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEK0350-BLK3)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEK0350-DUP1)		**Source: 23K0483-01**			Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	3.23		0.01	mmhos/cm		3.27			1.28	10
pH	7.4		1.0	units		7.4			0.269	10
Electrical Conductivity umhos	3230		10.0	umhos/cm		3270			1.28	10
Duplicate (BEK0350-DUP2)		**Source: 23K0518-04**			Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	0.68		0.01	mmhos/cm		0.68			0.337	10
Electrical Conductivity umhos	685		10.0	umhos/cm		682			0.337	10
pH	7.8		1.0	units		7.8			0.256	10
Reference (BEK0350-SRM1)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	576			umhos/cm	538.0		107	90-110		
Reference (BEK0350-SRM2)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0350-SRM3)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0350-SRM4)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0350-SRM5)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		

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Received: 11/09/2023 16:20
Reported: 11/16/2023 11:17

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0350 (Continued)										
Reference (BEK0350-SRM5)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEK0350-SRM6)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0350-SRM7)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0350-SRM8)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/09/23 16:20 23K0483

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 1 No. of Bottles: 1

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT AJR Farms/Maria C. Silveira
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Domestic Well	11/8/2023	1600	-0.6	ND
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/8/2023 @ 1600	11/9/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/9/2023 @9am	11-9-23 3:55
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/09/23 16:20 23K0483

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice:☐ Blue Ice:☐ None:☑

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)										
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										